

Lesson 3: Perform Testing and Documentation

LEARNING OBJECTIVES

At the end of the lesson, the student should be able to:

- Explain the steps involved in testing server functionality.
- Perform hands-on testing and create detailed server setup documentation, including configurations and test results.
- Reflect on the proper testing and documentation before deployment by recognizing how these practices improve consistency and productivity in everyday tasks.

INTRODUCTION

In this session, the learner will learn how to test if the computer server is working properly by understanding the safety precautions and guidelines for testing a configured computer server, recording and practicing hands-on testing, and recording important details such as test results and configurations. The learners are expected to perform testing based on their prior knowledge of the configuration process.

BACKGROUND AND Prerequisite:

You should already have prior knowledge about:

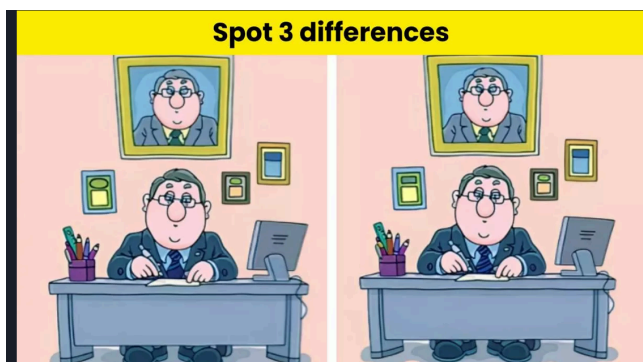
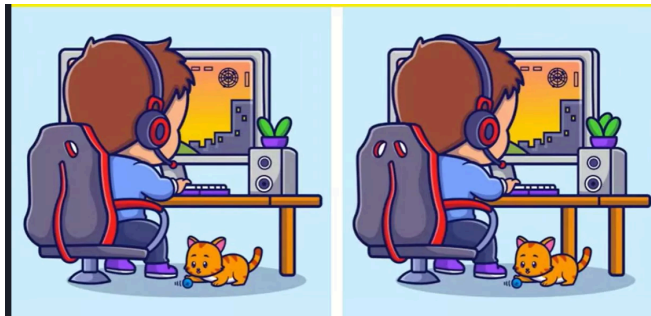
- The Tools and Requirements for Configuration
- Step-by-Step Configuration Process
- Security Considerations

LET'S TRY (INTRODUCTORY ACTIVITY)

TITLE: Spot the Difference

Instructions: Identify the difference in each picture representing “bugs” hiding from identical images.

SPOT 3 differences (BUG) for each picture.



(CONTENT:) LET'S EXPLORE

SAFETY PRECAUTIONS

Occupational Health and Safety (OHS) Policy – An Information and Communication Technology (ICT) student should know how to behave when working in the computer laboratory, and to accomplish tasks safely. Safety practices should be learned early and always adhered to when working with any electrical and electronic device, including personal computers and its peripherals. This is for your protection as well as for the people working with you, and for the devices that you are using.

Occupational safety and health (OSH) - is a planned system of working to prevent illness and injury where you work by recognizing and identifying hazards and risks. Health and safety procedures are the responsibility of all persons in the computer and technology industries. You must identify the hazards where you are working and decide how dangerous they are. Eliminate the hazard or modify the risk that it presents.

Occupational Health and Safety standards

Each student has a responsibility to their colleagues and their organization to report and act upon any potential workplace hazard. Students need to be aware of the type of hazards that are possibly present in their work environment.

Procedure

1. Identify the hazard.
2. Clear the area close to the hazard.
3. Partition the hazard off or clearly identify the area to protect people from harm.
4. If the hazard can be easily removed or cleared, then do so.

If not...

5. Report the hazard to the appropriate person (such as teacher, technician, etc.) to obtain assistance.
6. After clearing the hazard, fill out the correct documentation to assist in identifying improved practices to reduce further incidences of hazards.

All hazards must be reported using an **Accidental Report form**. This enables us to track the risks and hazards we have in our workplace and take action when necessary to make it safer for all students and clients.

Accident Reports

Accident forms are used to give specific details with regard to accidents that happened in the laboratory during experiments.

Accident reports contain the following details:

- ☐ Name of the person injured
- ☐ Date and time of the accident
- ☐ Injury sustained
- ☐ First aid given
- ☐ Action taken to prevent further accidents

Accident Report Sample Form

Form No:	Accident Report Form	Date:
		RM No:
Name:		Yr/Sec:
Type of Injury	Cause of Injury	Remedy

GUIDELINES FOR TESTING CONFIGURED COMPUTERS

Instruction: Read the following steps per part of the computer configuration and follow accordingly.

1. Preparing for the Test

Make sure of the following before beginning the testing process:

- Verify that the necessary hardware is installed and connected properly.
- Make sure that the operating system and all necessary programs are installed appropriately.
- Ensure that power sources and backup plans are in place in case of unexpected interruptions.

2. Hardware Testing

Test the physical components to guarantee proper operation:

- **CPU & RAM:** Check for overheating and stability using stress-testing tools.
- **Storage Drives:** Check the performance and condition of HDDs and SSDs by running disk inspections.
- **Graphics Card:** Test the stability and rendering performance using benchmarking tools.

3. Software Testing

Ensure that the installed software functions correctly:

- **Operating System:** Verify that the OS boots correctly and updates are installed.
- **Drivers:** Confirm all hardware drivers are up-to-date and correctly installed.
- **Security Software:** Ensure antivirus and firewall settings are active and properly configured.

4. Network & Connectivity Testing

- Verify that wired and wireless network connections are stable and functioning.
- Check internet speed and connectivity using testing tools.
- Check the workings of Bluetooth and other wireless peripherals.

Checking Client Requirements

Steps and Procedures:

1. Review Requirements: Go through the client's document carefully.
2. Ask for Clarification: Meet with the client to confirm unclear points.
3. Create a Checklist: List all confirmed requirements.
4. Map Features: Match requirements to system features.
5. Update the Document: Record any changes.
6. Get Approval: Ask the client to confirm the updated requirements.
7. Ongoing Checks: Regularly review requirements during development.

Importance of Ping Test.

- A ping test is crucial for network troubleshooting and monitoring because it verifies network connectivity
- Open Command Prompt by pressing Win + R, typing cmd, and pressing Enter.
- Type `ping [IP address or website]` (e.g., `ping 8.8.8.8` or `ping www.google.com`).
- Press Enter to start the test.
- Check the results for response time, packet loss, or errors.
- Type `exit` and press Enter to close the Command Prompt.

LET'S ANALYZE (Activity)

Instructional Guide:

1. Find and open the command line tool on your computer.
2. Type the Ping Command
3. Check the Results

Look for:

Reply: Confirms the server is reachable.

Time: Shows response speed.

Packet Loss: Indicates connection quality.

LET'S APPLY (Activity)

Instructional Guide: Perform a simulation to test different system components based on the specific requirements of a client or enterprise. Choose a known company or enterprise for your simulation. Focus on evaluating the following aspects, and explain your answer and how you have concluded that answer.

Example Aspects:

1. Hardware Specifications
2. Network Configurations
3. Security Measures

RUBRICS

Rating/Grade	Characteristics
Excellent (17-20 points)	Identifies all client requirements and conducts a thorough evaluation of hardware, network configurations, and security measures. Provides detailed, well-reasoned solutions backed by technical knowledge. The simulation is well-documented and accurately reflects real-world scenarios.
Proficient (13-16points)	Identifies most client requirements and assesses key system components with good accuracy, provides very good answers, although there can be a few details missing. The simulation is mostly realistic and complete.
Developing (9-12points)	Identifies some requirements but overlooks important aspects of the assessment. Some system components are evaluated, but the analysis lacks depth. Solutions may be incomplete, unclear, or not fully relevant.
Needs Improvement (5-8 points)	Shows minimal understanding of the client's requirements. The simulation is incomplete, lacks accuracy, and provides little to no relevant solutions. Major revisions are needed.
Score : /20	

LET'S ANALYZE (Short quiz)

Instructions: Answer the following questions based on the steps and procedures demonstrated in Ping Testing and Testing Simulation.

1. What does OHS stand for?

- a) Occupational Hazard Safety
- b) Occupational Health and Security
- c) Occupational Health and Safety
- d) Operational Hazard Standards

2. What is the main purpose of OHS policy in an ICT laboratory?

- a) To improve computer performance
- b) To enhance learning resources
- c) To ensure safety for students, devices, and colleagues
- d) To promote teamwork and cooperation

3. What is the first step in handling a workplace hazard?

- a. Report it immediately
- b. Ignore the hazard if it's minor
- c. Identify the hazard
- d) Evacuate the area

4. What should you do after identifying a hazard that cannot be cleared easily?

- a. Leave it for others to handle
- b. Report it to the appropriate person
- c) Ignore it if it seems minor
- d. Relocate all devices to another area

5. What is used to report accidents in the laboratory?

- a) Safety Checklist Form
- b) Hazard Warning Form
- c) Accident Report Form
- d) Equipment Inspection Form

6. Which of the following is NOT included in an Accident Report Form?

- a. Date and time of accident
- b. Type of injury sustained
- c. Number of students in class
- d. Action taken to prevent further accidents

7. How do you open the Command Prompt on Windows?

- a. Press Ctrl + Alt + Delete
- b. Type "cmd" in the search bar and press Enter
- c. Open the Task Manager and select "Run New Task"
- d. Use the Windows PowerShell

8. What command do you use to ping Google's website?

- a. test google.com
- b. ping google.com
- c. connect google.com
- d. trace google.com

9. What information can you check from the ping test results?

- a. Response time and packet loss
- b. Internet download speed
- c. List of connected devices
- d. Network password

10. How do you perform a continuous ping on Windows?

- a. ping -c google.com
- b. ping -t google.com
- c. ping google.com
- d. ping -p google.com

11. Why is a ping test important?

- a) It speeds up your computer
- b) It verifies network connectivity
- c) It installs essential updates
- d) It prevents data loss

12. What is the correct command to initiate a ping test in Command Prompt?

- a) testping 8.8.8.8
- b) checkping google.com
- c) ping 8.8.8.8
- d) startping google.com

13. What indicates a successful ping test?

- a) Error messages in the results
- b) Long response times
- c) Stable response time and no packet loss

d) Sudden disconnection from the network

14. Which component is tested for overheating and stability during hardware testing?

a) Graphics Card

b) Network Adapter

c) CPU & RAM

d) Power Supply

15. Which of the following is checked to ensure stable internet connectivity?

a) Disk inspection tool

b) Bluetooth settings

c) Internet speed test

d) Antivirus software

LET'S REFLECT

INSTRUCTION: Answer this question, what is the importance of completing proper testing before documentation?

ANSWER KEY:

1. c
2. c
3. c
4. b
5. c
6. c
7. b
8. b
9. a

10. B

11. B

12. C

13. C

14. C

15. c